LESSON 6.12

Least Common Multiple

Start here · Words, worked example, and sentence starters

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can find the least common multiple (LCM) of two numbers by listing or comparing their multiples.

WHY THIS MATTERS Astronaut Reyes checks the oxygen system every 4 hours. Astronaut Chen checks the water recycler every 6 hours. They both start their shifts at 6:00 AM. When is the next time both astronauts will be checking their systems at the same time?

START HERE

Words for this lesson

Read each word before you begin. Say it out loud.

WORD	WHAT IT MEANS	EXAMPLE
Least Common Multiple *Spanish: Mínimo común múltiplo*	The least positive number that is a multiple of two or more numbers.	Multiples of 4 and 6 first meet at 12, so LCM = 12.
Multiple *Spanish: Múltiplo*	What you get when you multiply a number by 1, 2, 3, and so on.	Multiples of 4: 4, 8, 12, 16, 20 — found by multiplying 4 × 1, 4 × 2, 4 × 3, ...
Common multiple *Spanish: Múltiplo común*	A number that two or more numbers both go into.	12, 24, and 36 are all common multiples of both 3 and 4
Skip counting *Spanish: Conteo salteado*	Counting by a number, like 2, 4, 6, to list its multiples.	Skip count by 6: 6, 12, 18, 24, 30 — each jump adds 6
Prime factorization *Spanish: Factorización prima*	Writing a number as prime numbers multiplied together. It helps you find the LCM.	4 = 2 × 2, 6 = 2 × 3. LCM uses the highest power of each prime: $2^2 \times 3 = 12$

HOW IT WORKS

Worked example

These numbers are not on your problems. The steps are. Follow them with your own numbers.

I want the LCM of 4 and 6. I skip count the multiples of 4: 4, 8, 12, 16, 20, 24.

1. Then I skip count the multiples of 6: 6, 12, 18, 24.
2. I look for the first number in BOTH lists. 12 is in both lists, and it comes before 24.
3. So the LCM of 4 and 6 is 12. I check: 12 ÷ 4 = 3 and 12 ÷ 6 = 2, so both go into 12 evenly.

NOW YOU TRY

Same steps, your turn

The steps are the same as the worked example. The numbers are yours. Do the work.

Let's find the LCM of 3 and 5. What are the multiples of 3? (3, 6, 9, 12, 15.)

1. _____
2. _____
3. _____

Answer: _____

SAY IT AND WRITE IT

Sentence starters

Finish each sentence out loud with a partner. Then use them in your writing.

- The LCM of 4 and 6 is _____ because it is the smallest number in the _____ group.

Word bank Least Common Multiple | Multiple | Common multiple | Skip counting | Prime factorization | least | common | multiple | smallest | 12 | shared | both

WATCH OUT!

A common mistake

A common mistake in Least Common Multiple is multiplying the two numbers together instead of listing multiples to find the smallest shared one. For example, for 4 and 6, a student calculates 4 × 6 = 24 and calls that the LCM, but the actual least common multiple is 12 (multiples of 4: 4, 8, 12

LESSON 6.12

Least Common Multiple

Version A · Supported practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can find the least common multiple (LCM) of two numbers by listing or comparing their multiples.

WHY THIS MATTERS Astronaut Reyes checks the oxygen system every 4 hours. Astronaut Chen checks the water recycler every 6 hours. They both start their shifts at 6:00 AM. When is the next time both astronauts will be checking their systems at the same time?

PART 1 Understand the idea

models and meaning

1 **Mark and label each value on the number line.**

Find the LCM of 5 and 6 on the number line



HINT Read the labels: some marks are multiples of 5 and some are multiples of 6. The LCM is the first spot that belongs to BOTH.

SHOW YOUR THINKING

PART 2 Apply it

real-world problems

2 **Circle the letter of the best answer. Show how you know.**

What is the LCM of 3 and 5?

(A) 3
(B) 5
(C) 15
(D) 30

HINT LCM is about MULTIPLES (counting UP by each number), not factors. You want the smallest number BOTH lists reach.

SHOW YOUR WORK

3 **Circle the letter of the best answer. Show how you know.**

What is the LCM of 6 and 8?

(A) 6
(B) 14
(C) 24
(D) 48

HINT Careful: 6 + 8 is not a multiple of either number. You need a number that both 6 AND 8 divide into evenly.

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

What is the LCM of 4 and 10?

(A) 2
(B) 10
(C) 20
(D) 40

HINT The LCM is the smallest number both 4 and 10 count into — it is often much smaller than 4 × 10 = 40.

SHOW YOUR WORK

5 Circle the letter of the best answer. Show how you know.

What is the LCM of 2 and 7?

(A) 2
(B) 7
(C) 9
(D) 14

HINT Notice that 2 and 7 are both prime — they share no common factor. That makes this a special, faster case.

SHOW YOUR WORK

PART 3 Explain and correct

reasoning

6 Find the mistake. Explain it, then show the correct work.

Spot Riley's Mistake — LCM of 4 and 6

1. **Problem** Find the LCM of 4 and 6.
2. **Multiples of 4** 4, 8, 12, 16
3. **Multiples of 6** 6, 12, 18
4. **Choose the answer** Riley added the two numbers: 4 + 6 = 10, so LCM = 10.

Which step has the mistake? Explain what went wrong, then show the correct work.

HINT Riley added 4 + 6 to get 10. Check: is 10 in both lists of multiples?

CORRECT WORK

Correct answer: _____

EXPLAIN YOUR THINKING

What is the least common multiple of 4 and 6? How did sorting the multiples help you find it?

SENTENCE STARTER The LCM of 4 and 6 is ___ because it is the smallest number in the ___ group.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help

LESSON 6.12

Least Common Multiple

Version B · Core practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can find the least common multiple (LCM) of two numbers by listing or comparing their multiples.

WHY THIS MATTERS Astronaut Reyes checks the oxygen system every 4 hours. Astronaut Chen checks the water recycler every 6 hours. They both start their shifts at 6:00 AM. When is the next time both astronauts will be checking their systems at the same time?

PART 1 Understand the idea

models and meaning

1 **Write the letter of the matching item on each line.**

Match each pair of numbers to their LCM.

_____ LCM(2, 5)	(A) 10
_____ LCM(4, 10)	(B) 20
_____ LCM(3, 8)	(C) 24
_____ LCM(6, 9)	(D) 18
_____ LCM(5, 7)	(E) 35
_____ LCM(4, 6)	(F) 12

2 **Complete the table. Show how you found each value.**

List multiples of each number until you find the LCM.

Numbers	Multiples of First	Multiples of Second	LCM
6 and 10			
8 and 12			
9 and 12			

SHOW YOUR WORK

3 **Write the name of the correct group on each line.**

Sort each pair: does the LCM equal the product of the two numbers, or is the LCM less than the product?

Groups: **LCM = Product (no shared factors)** **LCM < Product (shared factors)**

_____ 3 and 7

_____ 5 and 11

_____ 2 and 9

_____ 4 and 6

_____ 6 and 8

_____ 10 and 15

PART 2 Apply it

real-world problems

4 **Circle the letter of the best answer. Show how you know.**

Two rovers leave the space station at 8:00 AM. Rover A returns every 15 minutes and Rover B returns every 20 minutes. When is the next time both rovers are at the station together?

(A) 8:30 AM (30 minutes later)

(B) 8:40 AM (40 minutes later)

(C) 9:00 AM (60 minutes later)

(D) 9:20 AM (80 minutes later)

SHOW YOUR WORK

PART 3 Explain and correct

reasoning

5 Find the mistake. Explain it, then show the correct work.

Spot Maya's Mistake — LCM of 6 and 9

1. **Problem** Find the LCM of 6 and 9.
2. **Multiples of 6** 6, 12, 18, 24
3. **Multiples of 9** 9, 18, 27
4. **Choose the answer** Maya saw that 9 is bigger than 6, so she wrote LCM = 9.

Which step has the mistake? Explain what went wrong, then show the correct work.

CORRECT WORK

Correct answer: _____

EXPLAIN YOUR THINKING

What is the least common multiple of 4 and 6? How did sorting the multiples help you find it?

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help

LESSON 6.12

Least Common Multiple

Challenge · Extension

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can find the least common multiple (LCM) of two numbers by listing or comparing their multiples.

WHY THIS MATTERS Astronaut Reyes checks the oxygen system every 4 hours. Astronaut Chen checks the water recycler every 6 hours. They both start their shifts at 6:00 AM. When is the next time both astronauts will be checking their systems at the same time?

PART 1 Understand the idea

models and meaning

1 **Complete the table. Show how you found each value.**

Use prime factorization to find the LCM. Take the highest power of each prime factor.

Numbers	Prime Factorizations	Highest Powers	LCM
12 and 18			
15 and 20			
8 and 14			

SHOW YOUR WORK

PART 2 Explain and correct

reasoning

2 Find the mistake. Explain it, then show the correct work.

Find Jaylen's Mistake

1. **Find LCM of 8 and 12** List multiples of 8 and 12
2. **Multiples of 8** 8, 16, 24, 32, 40, 48
3. **Multiples of 12** 12, 24, 36, 48
4. **Common multiples** 24, 48
5. **Choose LCM** LCM = 48

Which step has the mistake? Explain what went wrong, then show the correct work.

CORRECT WORK

Correct answer: _____

3 Answer in complete sentences.

You are making party favors. Glow sticks come in packs of 10 and bracelets come in packs of 4. What is the fewest of each you can buy so every favor has exactly one glow stick and one bracelet with none left over? Explain how the LCM solves this problem.

WRITE YOUR ANSWER

EXPLAIN YOUR THINKING

What is the least common multiple of 4 and 6? How did sorting the multiples help you find it?

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help